DEBARGHYA SENGUPTA

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PROFILE

Full-stack software engineer with hands-on experience building production-grade AI systems — RAG pipelines, fraud-detection engines, and OCR-powered automation. Proficient in Java (DSA), Python, C++, and C with strong OOP and system design fundamentals. Comfortable across the full stack (Next.js, FastAPI, Node.js, PostgreSQL) and AI layers (LLMs, embeddings, YOLO, EasyOCR). Driven by scalability, zero-cost constraints, and real-world usability.

PROJECTS

Fin-Sight — Financial RAG Platform *2024*

Next.js 15 · FastAPI · PostgreSQL · MiniLM · Groq Llama-3.3-70B · Celery · Presidio · FinBERT · Docker · Git

- • Built a multi-tenant RAG platform for SEC 10-K filings — cited Q&A, side-by-side comparison, anomaly alerts, and live EDGAR watch — with GitHub Actions CI/CD pipelines, running on $0/month dev stack.
- Designed a zero-extension vector store using JSON embeddings in PostgreSQL + NumPy cosine search, eliminating Qdrant/Pinecone while keeping the escape hatch to pgvector HNSW a one-column migration.
- Implemented adaptive chunking (800-char prose windows + whole-table preservation) that reduced half-table citation errors from ~30% to near-zero on a 200-pair hand-graded eval set.
- Achieved <660 ms P95 end-to-end query latency (embed → cosine → cross-encoder rerank → Groq Llama) on CPU; ~6.6× cheaper than equivalent GPT-4o + Pinecone stack at 1,000 users (~$386 vs $2,563/month).

Nyatik Nayan — Retail Fraud Detection Platform *HackArena 2.0*

React 19 · Node.js/Express · FastAPI · PostgreSQL · Redis · YOLOv10 · EasyOCR · RapidFuzz · Groq Llama 3.1 · Git

- • Ranked Top 10 in HackArena 2.0 Zonal Round; built real-time barcode fraud detection using YOLOv10, EasyOCR, Redis, and Groq Llama.
- Built a real-time barcode fraud engine processing scans in <1 second: inventory cross-reference, scan-frequency analysis, barcode-age heuristics, MK ID validation, and a decision-tree risk scorer.
- Integrated YOLOv10 object detection + EasyOCR + RapidFuzz fuzzy matching for image-based label-swap detection; transactions with trust score <65% blocked automatically with Llama-generated fraud narratives.
- Engineered Redis-backed UID uniqueness gate and NX transaction locks to prevent double-scanning, with WebSocket push for instant checkout UI updates and SendGrid escalation on 3+ flags within 24 hours.

MediHub — AI Prescription Manager *March 2026*

Node.js · Express · PostgreSQL · Flask · REST APIs · MobileNetV2 (CNN) · EasyOCR · KNN (Scikit-Learn) · Nodemailer · Node-Cron · Git

- • Developed end-to-end AI prescription digitization pipeline using CNN (MobileNetV2), EasyOCR, KNN classification, and automated reminders with Node-Cron.
- Built automated appointment reminder system using Node-Cron + Nodemailer — daily DB scan sends personalized 24-hour pre-visit email alerts, reducing missed follow-ups.
- Delivered secure multi-user platform (Bcrypt + Express-Session) with RESTful API backend and a real-time health dashboard showing active prescriptions and unique-doctor counts; sole developer from concept to deployment.

TECHNICAL SKILLS

Languages: Java (DSA), Python, C++, C, JavaScript, TypeScript, SQL | **Frameworks:** Next.js, FastAPI, Express.js, Flask, React
AI / ML: LLM APIs (Groq, OpenAI), RAG pipelines, Sentence Transformers, YOLOv8/v10, EasyOCR, FinBERT, Presidio, Scikit-Learn, TensorFlow
Infrastructure: PostgreSQL, Redis, Celery, Docker, Git, AWS (S3/SES/DynamoDB), Clerk Auth, SendGrid, Node-Cron, Alembic, pytest
Concepts: Data Structures & Algorithms, OOP, System Design, REST APIs, Microservices, Distributed Systems, Scalability, Agile/Scrum

LEADERSHIP & EXTRACURRICULARS

Governor — Mahakash Club (Space Club, GNIT) *Present*

- • Leading GNIT's space & astronomy club — organizing workshops, stargazing, and outreach programs.

Cloud Computing Head — Microsoft Student Associates (MSA), GNIT *2026*

- • Leading Cloud Computing for MSA chapter — Azure sessions, cert prep, and hands-on labs.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Developer Professional — Oracle
- OCI 2025 Certified Generative AI Professional — Oracle
- Oracle AI Autonomous Database 2025 Certified Professional — Oracle
- Get Started with Azure Management Tasks — Microsoft Learn
- Fundamentals of Cybersecurity — Zscaler

EDUCATION

B.Tech / B.E. in Computer Science [Guru Nanak Institute of Technology], [Kolkata]

2023 – 2027 (Expected) | CGPA: 8.3 (through 5th Semester)